Analyzing Bulk RNA-Seq Data from BMP7-Treated Medulloblastoma Cells

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BCB420

Assignment 3: Pathway and Network Analysis Presentation

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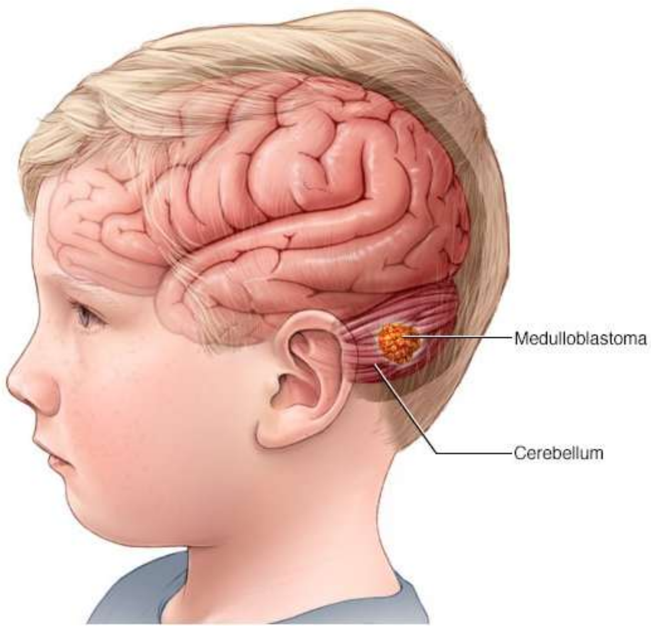
Medulloblastoma (MB) is a Pediatric Brain Cancer

- Rare and Aggressive; 5 cases per 1 million individuals.
- Originates in cerebellar progenitor cells.
- Bone Morphogenetic protein (BMP) is critical in MB development but has an unknown mechanistic role.

- Ohata et al. aimed to **explore how BMP pathways contribute to the development of medulloblastoma.**



https://www.mayoclinic.org/diseases-conditions/medulloblastoma/symptoms-causes/syc-20579268

Experimental Design

- **4** medulloblastoma cell lines: CHLA01, CHLA01R, MB002, and D425
- **2** conditions: PBS (Control) and BMP7, each treated for 72 hours
- **3** replicates per cell line per condition
- Total 24 samples
- Bulk RNA sequencing using Illumina NovaSeq
- GEO Accession ID: GSE229150

BMP7 Signaling Promotes Tumor Survival

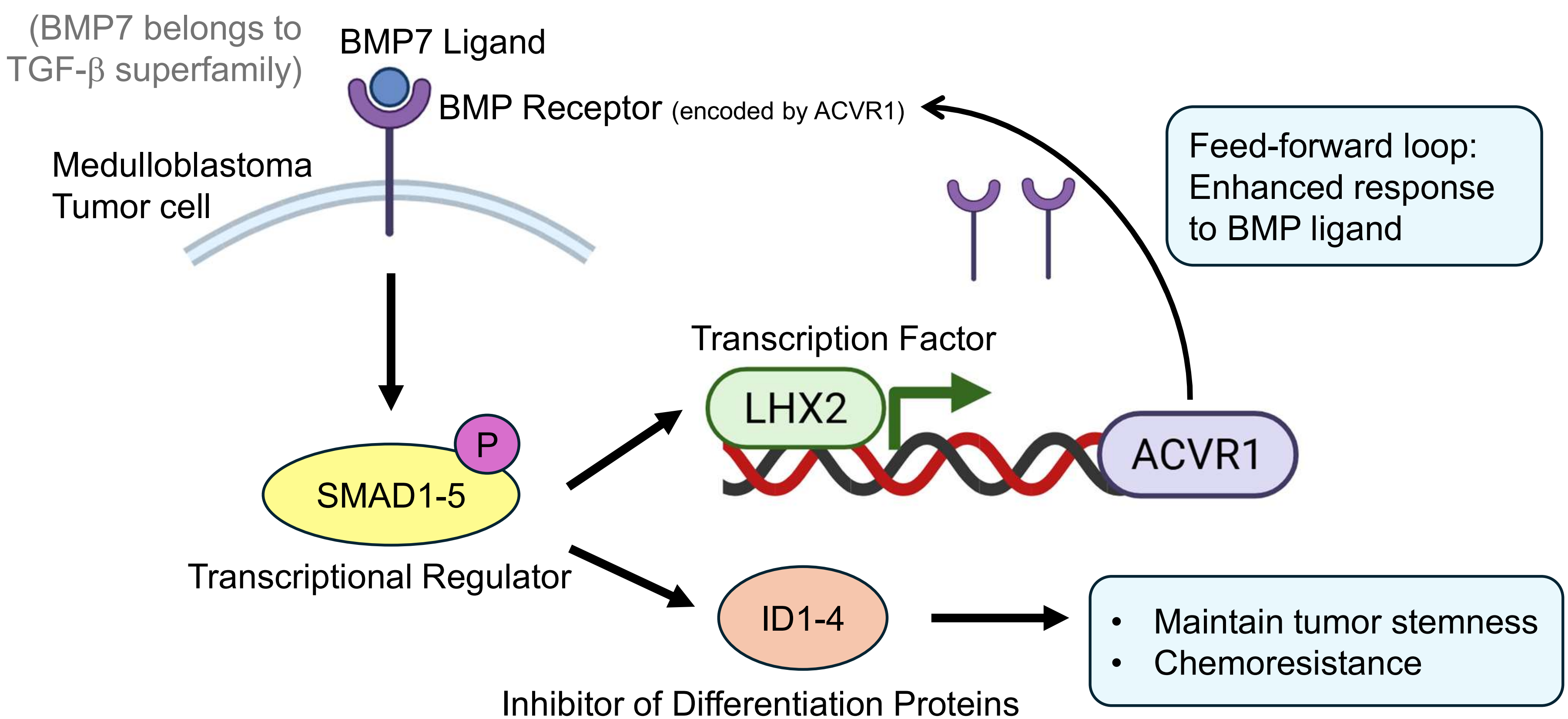


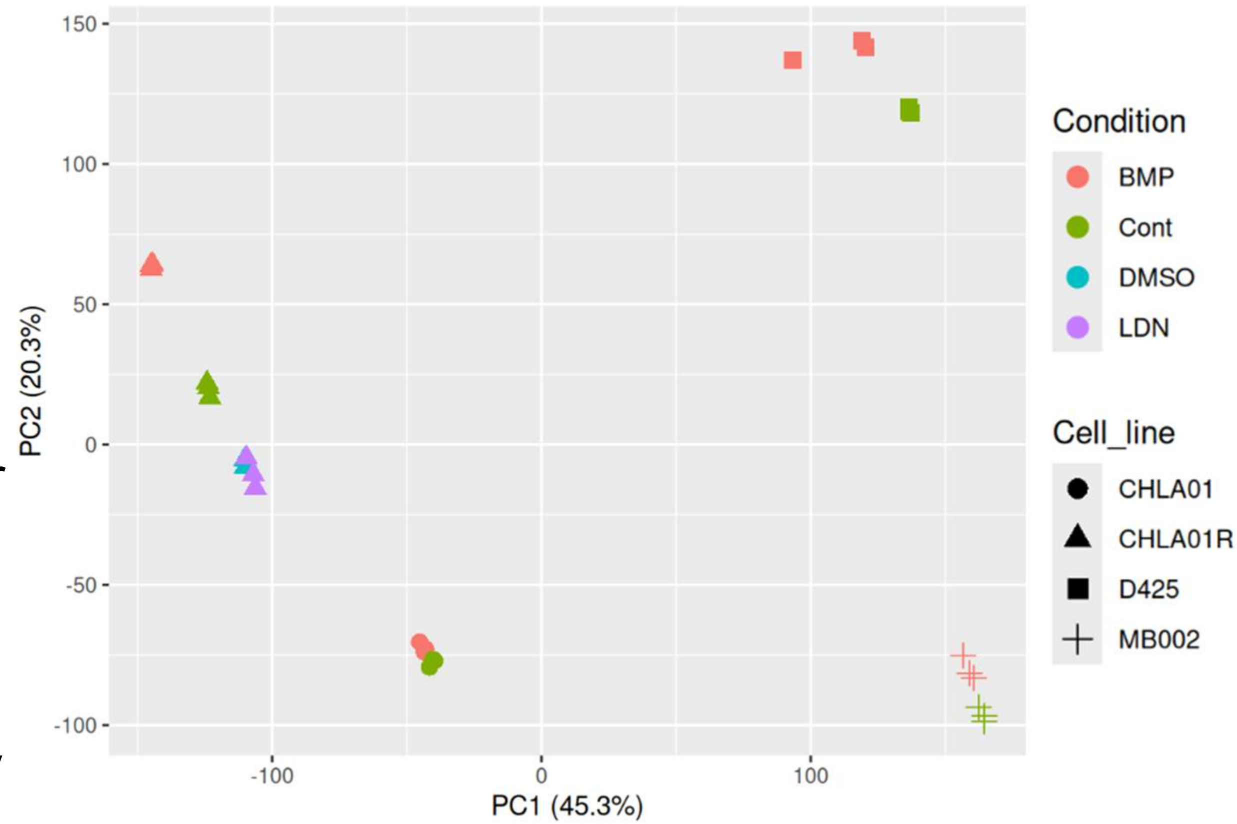
Figure Created Using bioRender

Data Cleaning

- Pre-filter: 66,023 genes
  After filtering out low read counts: 17,462
- <u>Sample Visualization:</u>
  - There are no clear outliers.
  - Clustering is mainly due to cell line, and within a cell line, samples tend to cluster by condition.

This clustering pattern is important for differential expression analysis, particularly for the model design.

PCA plot of log-CPM gene expression

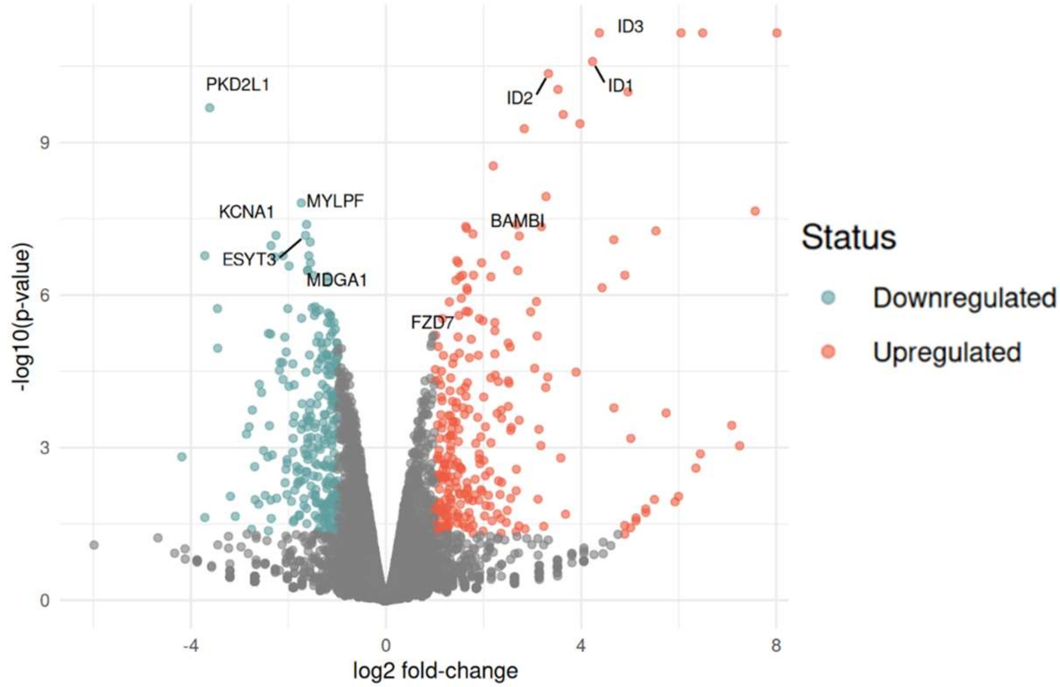


Data Normalization and DE Analysis

- Normalization method: **Trimmed Mean of M-values (edgeR package)**
  - Assumes most genes are not differentially expressed.
  - Accounts for library size and RNA composition differences between samples.

- Differential Expression Analysis **(edgeR package)**
  - Tagwise differential expression generated by the **quasi-likelihood (QL)** test.
  - Reference/Baseline = PBS treated Control
  - Focused on CHLA01 cell line (3 samples per condition) as it showed tight clustering of BMP samples and control samples in the PCA plot.

BMP7 vs Control in CHLA01 cell line



|log2FC| > 1 and p.adj < 0.05
(Adjusted p-value based on Benjamini-Hochberg method)

Methods: Non-thresholded analysis using GSEA

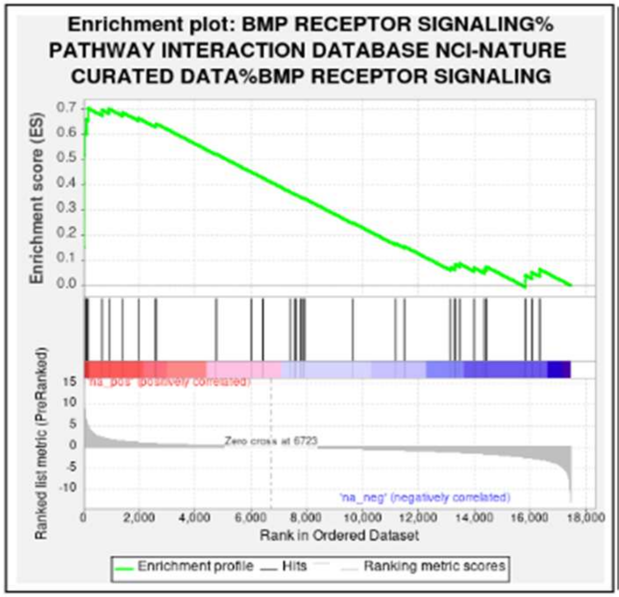
- Gene Set Enrichment Analysis (GSEA)
  - Reason: Does not requiring an arbitrary significance threshold, enabling detection of coordinated but modest expression changes across biological pathways. GSEA desktop interface allows interactive exploration to test and refine analysis

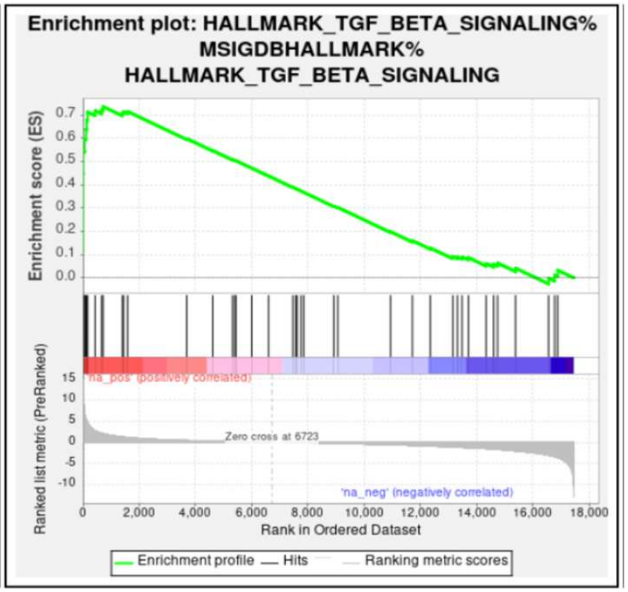
  Gene Set: Custom GMT gene-set file curated by the Bader Lab
  - Reason: Integrates pathway and functional annotations from multiple databases, Updated regularly.
  - Human_GOBP_AllPathways_noPFOCR_no_GO_iea_September_01_2025_symbol.gmt corresponding to the September 01, 2025.
  - GMT gene set file was filtered to include only pathways having 15 -100 genes.

- Ranked gene list: Genes were ordered by -log10(Pvalue) *sign(logFC)
  - The analysis was run with 1,000 permutations.

Enriched Pathway Results

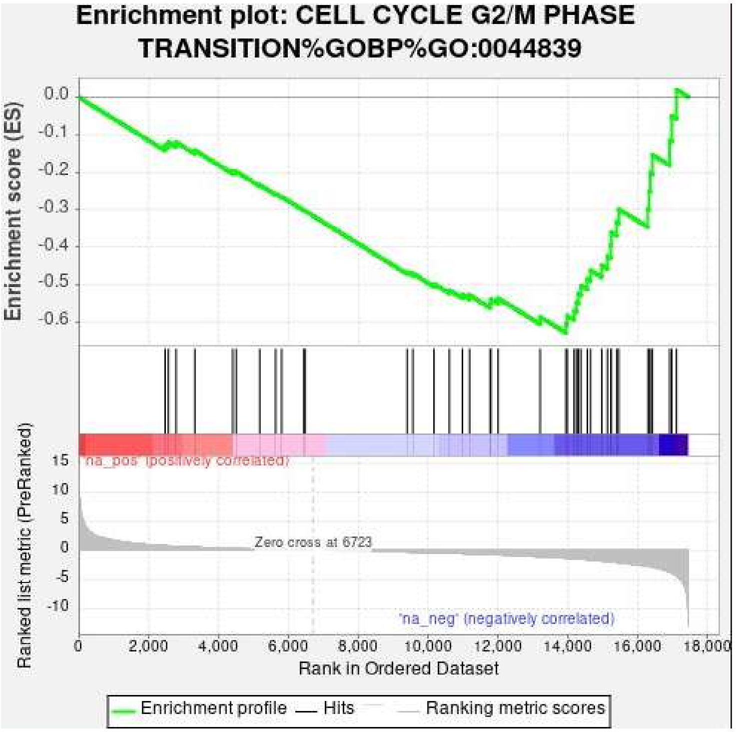
1. 881 gene sets showed positive enrichment scores.
   - 364 gene sets achieved statistical significance at FDR < 25%.
   - 153 gene sets reaching high significance at a nominal p-value < 1%.
2. Top enriched pathways with positive enrichment score:
   - BMP Receptor Signaling
   - TGF-β signaling



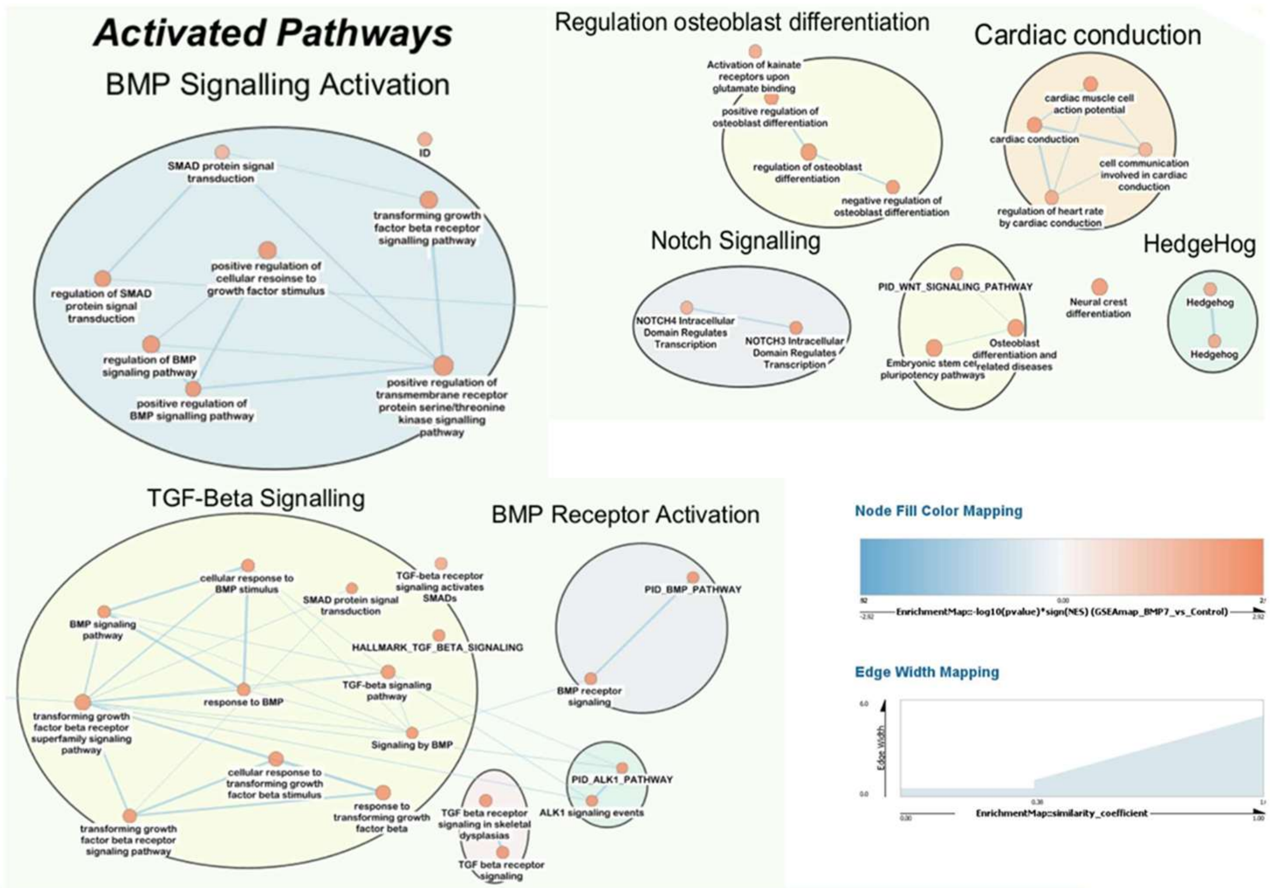


Enriched Pathway Results

1. 4,086 gene sets showed negative enrichment scores.
   - 414 gene sets achieved statistical significance at FDR < 25%.
   - 220 gene sets reaching high significance at a nominal p-value <1%.
2. Top enriched pathways with negative enrichment score:
   - Cell Cycle G2/M Phase Transition
   - Microtubule Cytoskeleton Formation
   - Negative Regulation of Macroautophagy

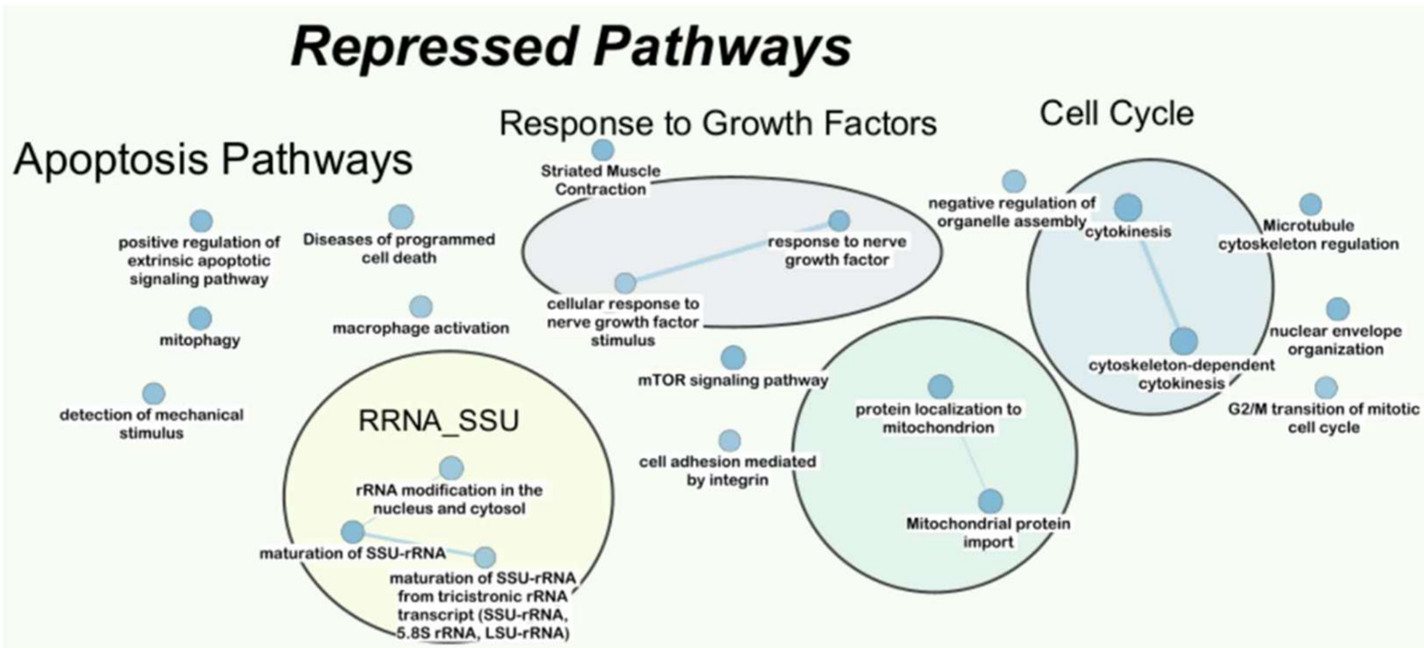


Cytoscape Visualization: Activated Pathways



FDR Q-value Cutoff: 0.2
p-value Cutoff 0.01

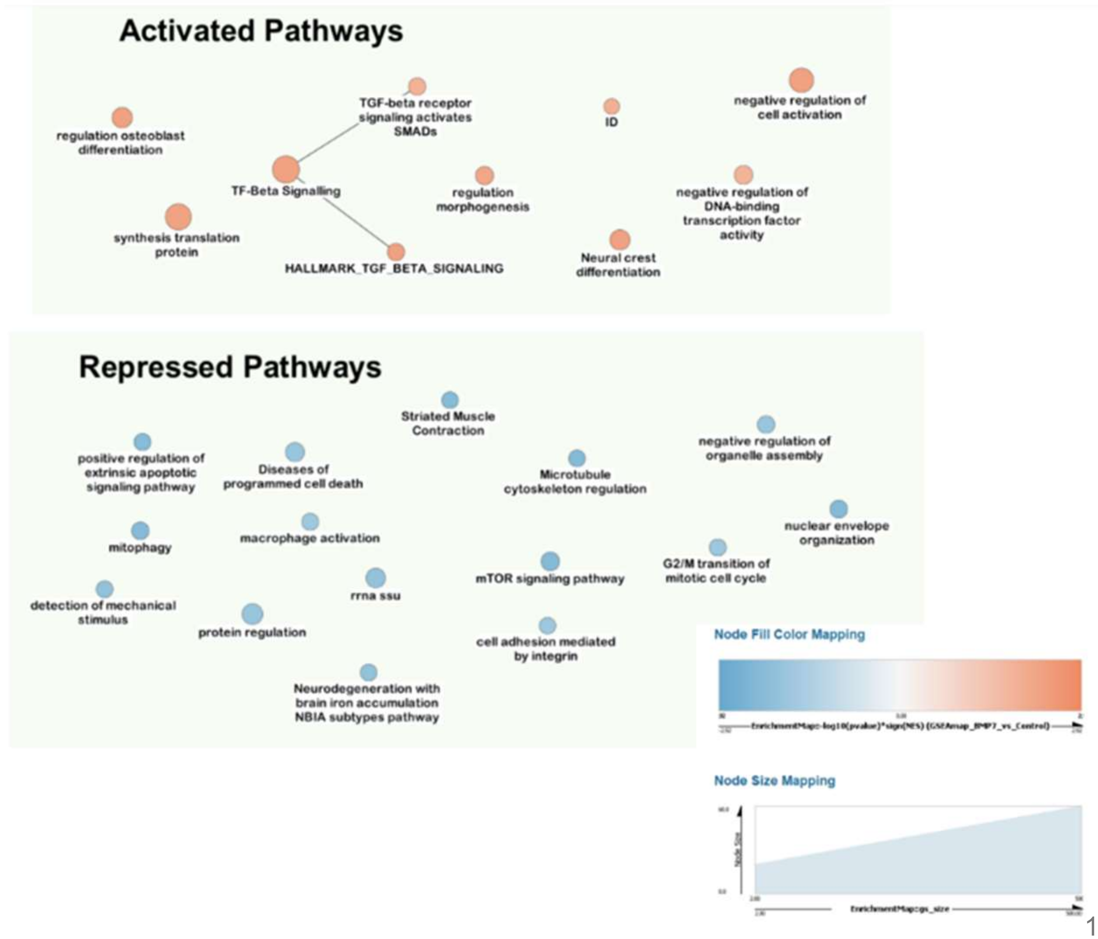
Cytoscape Visualization: Repressed Pathways



FDR Q-value Cutoff: 0.2
p-value Cutoff 0.01

Results: Major themes

1. TGF-Beta Signaling, ID, SMAD regulation are activated and fits the model of **oncogenic BMP signaling** (Ohata et al. 2025).
2. Apoptotic Pathways and Cell Cycle Pathways are repressed pathways which fits the model that **suppressing apoptosis and cell proliferation checkpoints promotes MB tumor growth** (Ohata et al. 2025).



Discussion and Key Insights

- **Support for Activated Pathways**
  - Activated pathway clusters corresponding to TGF-beta/BMP signaling, BMP receptor Activation, SMAD activation and ID protein signaling, reflecting active BMP/SMAD signaling in BMP7-treated medulloblastoma cells.

- **Support for Repressed Pathways**
  - Cell cycle progression is one of the repressed pathways in the analysis, supporting the paper.
  - While apoptotic pathways and response to growth factors are not explicitly mentioned in the paper, they can be supported by the fact that MB tumors rely on the suppression of apoptosis and growth factor sensitivity to maintain tumor cell survival and cancer stem cell populations.

Thresholded Over-representation Analysis (ORA)

- Genes filtered based on p-value < 0.05 and log fold change > 1 (up-regulated) and p-value < 0.05 and log fold change < -1 (down-regulated)
  - Number of up-regulated genes: 376
  - Number of down-regulated genes: 352
  - Number of combined-regulated genes: 728
- Custom GMT gene-set file curated by the Bader Lab same as for non-thresholded analysis. Pre-filtered to include only pathways having 15 -100 genes.
- **g:Profiler** package used (widely used, regularly updated, multiple databases)
  - Applied FDR correction for multiple hypothesis testing

Comparison: Thresholded vs Non-thresholded

- Pathways common in thresholded and non-thresholded analysis for up-regulated genes: HALLMARK_TGF_BETA_SIGNALING, BMP RECEPTOR SIGNALING
- Pathways common in thresholded and non-thresholded analysis for down-regulated genes: CELLULAR COMPONENT ASSEMBLY INVOLVED IN MORPHOGENESIS, INTEGRIN CELL SURFACE INTERACTIONS
- Notable differences in the pathways returned by each method.
- Thresholded ORA analysis on upregulated genes returned pathways associated with terms like “CARDIAC”, “AORTIC” and “ARTERY”, as opposed to mechanistic pathways like TGF-Beta, BMP, ID, mentioned in the paper and observed in our EnrichmentMap.
- Similarly, for downregulated genes, the thresholded analysis returned pathways associated with terms like “MYOFIBRIL” and “MUSCLE CELL”, as opposed to the cell cycle and apoptosis pathways.
- This discrepancy is likely due to the fundamental methodological difference between the two approaches.
  - ORA relies on genes that pass a strict significance threshold. Genes which are subtly but consistently upregulated, may not have enough strongly significant genes to dominate the ORA results.
  - GSEA considers the entire ranked gene list, making it more sensitive to detecting coordinated transcriptional shifts across a pathway, even when individual gene changes are modest.

- **This explains why GSEA more clearly recovers the BMP/SMAD/TGF-Beta signaling themes that are central to the paper’s proposed mechanism, and why it is considered a more powerful method for pathway analysis in this type of experiment.**

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